

Luis A. Silva LLanca

800 Lancaster Avenue

Villanova, PA 19085

telephone: 610 – 514 – 7797

e-mail: luis.silva@villanova.edu



CURRICULUM VITAE

EDUCATION

Universidad de Santiago de Chile

M.S. Mechanical Engineering (2004 – 2006)

Dissertation: “Fluid Mechanics and Heat Transfer in Non Newtonian Fluid Including Phase Change”

Advisor: Dr. Nelson Moraga

Universidad de Santiago de Chile

B.S. Mechanical Engineering (1999 – 2004)

TEACHING EXPERIENCE

Universidad de Santiago de Chile

Assistant Professor: Machine Elements Design (Fall 03 – Spring 05)

Universidad de Santiago de Chile

Assistant Professor: Finite Volume Method (Spring 04 – Fall 05)

Universidad de Santiago de Chile

Part-time Professor: Mathematical and Computational Modeling (Spring 07)

PROFESSIONAL AFFILIATIONS

CADE-IDEPE Internship (Engineering Consultant Company)

Santiago, Chile (Winter 04)

PUBLICATIONS

Articles

Moraga, N., Silva, L., “Forced Convection of Non Newtonian Fluid (Ostwald de Wale) Inside Ducts”, in *XVI Congreso de Ingeniería Química CIQ 2005 (Chemical Engineering Workshop)*; Pucón ,Chile (November 2005).

Moraga, N., Silva, L., “Natural Convection Heat Transfer in Sterilization of Non Newtonian Liquid Food”, in *IV Jornada de Ingeniería Mecánica Computacional (Computational Mechanical Engineering Workshop)*; Universidad de Concepción, Chile (September 2006).

Moraga, N., Silva, L., Ortega, A., “Unsteady Natural Convection of a Canned Non-Newtonian Liquid Food”, in *Proceedings of 2009 ASME Summer Heat Transfer Conference*, San Francisco, CA, (July 2009).

Silva, L., Moraga, N., Ortega, A., “Unsteady Buoyant Convection and Liquid-Solid Phase Change in a Power Law Fluid Within a Differentially Heated Square Cavity”, in *Proceedings of 2009 ASME International Mechanical Engineering Congress and Exposition*, Orlando, FL, (November 2009).

Silva, L., Ortega, A., “Convective Heat Transfer Due To an Impinging Synthetic Jet: A Numerical Investigation of a Canonical Geometry”, in *Proceedings 12th IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electric Systems*, Las Vegas, NV, (May 2010).

Silva, L., Ortega, A., “Numerical Simulation of Local Heat Transfer and Scaling of a Synthetic Impinging Jet in a Canonical Geometry”, in *Proceedings of the ASME 2010 International Mechanical Engineering Congress & Exposition*, Vancouver, British Columbia, Canada, (November 2010).

Reviews

Review of **Efficiency Modeling Of Hybrid Photovoltaic Thermal Systems**, ASME 2nd International Conference on Energy Sustainability, August 2008.

Review of **Numerical Simulation Of Unsteady Mixed Convection In A Minichannel With Multiblocks Under Imposed Flow Modulation**, The Sixth International Conference on Nanochannels, Microchannels, and Minichannels, June 2008.
